Linear Algebra Workshop

Little Hall Tutorial Program

Semester 2, 2026

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How to prepare

1. Make a4 notes sheet draft.
   1.1 Do a run through your notes.
   1.2 Linear Algebra is all patterns.
2. Look back at difficult tutorial questions.
3. Try past exam questions.

Be careful:

1. Don't waste your time.
   Linear algebra is all about calculations.
2. Timed practice exams can burn you out.

Standard forms in Linear Algebra

You should go through and make your own table of these standard forms and include some on your a4 notes sheet.

1. Find the inverse of this matrix
2. Gram-Schmidt procedure
3. Are these vector linearly (in)dependent?
4. Diagonalise this matrix
5. What is the change of basis matrix
6. Cofactor expand to calculate a determinant
7. Linear regression work (calculate the line of best fit, etc.)

   ... and so on.

What goes in your a4 notes? — Examples

1. 'If ... then ...' (Linear Algebra is pattern recognition).
   1.1 If I'm asked to invert a matrix I need to do

$$[A|I_n] \sim \cdots \sim [R|B].$$

   1.2 If I'm asked to diagonalise a matrix I need to do the steps ....
   1.3 Gram-Schmidt procedure.
2. Determinant formulas.
3. Volume of parallelepiped

$$\text{Volume} = |u \cdot (v \times w)|.$$

4. Some basic results that make answering questions fast.
   4.1 e.g. if $k > n$ and you have k vectors $\{v_1, \ldots, v_k\}$ in an n-dimensional space V then the vectors are linearly dependent.
   4.2 Orthogonal matrices have $\det Q = \pm 1$.
   4.3 Trace is the sum of eigenvalues, determint is the product of eigenvalues.

Advice for the a4 notes

1. If you didn't use it in tutorials, it's probably too specific for the exam.
2. Don't write down everything (this is why you make a draft).
3. Include 'safety facts'. I liked putting the inverse of a 2×2 matrix, even though I know it by heart.

Exam strategies

1. Look for easy marks at the start. e.g. invert a matrix question.
2. Skipping around is great.
   2.1 Use your formula sheet to mark skips.
3. Understand who is marking your exam.
4. Draw pictures where possible.
5. The last 5 minutes:
   5.1 Write down key words or theorems.
   'Use rank-nullity theorem', 'the vectors are linearly dependent'.
   5.2 Check answers are in the form required. e.g. when working in

$$\mathcal{P}_3 = \{a_0 + a_1 x + a_2 x^2 + a_3 x^3\}$$

vectors should (ideally) be written

$$1 + 2x^2 + x^3 \text{ and not } \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}.$$

Open questions

Where can I get these slides?

